

# Junqiao Fan

Ph.D. Candidate (thesis submitted) | Nanyang Technological University, Singapore  
School of Electrical and Electronic Engineering | Expected graduation: December 2026

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Open to postdoctoral opportunities.

## Research Interests

**Multimodal human perception:** Privacy-preserving mmWave radar sensing; human motion capture and human pose/mesh estimation.

**Generative AI:** Human motion generation and modeling; diffusion and flow models.

**Robotics:** Robot learning; multimodal human-robot interaction; human-centered embodied intelligence.

## Education & Research Experience

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|----------------------------------------------------------------------------------------------------------------------------|--------------|
| Nanyang Technological University, Singapore                                                                                | 2022–Present |
| Ph.D., School of Electrical and Electronic Engineering.<br>Supervisor: Prof. Lihua Xie; Co-supervisor: Prof. Jianfei Yang. |              |
| The Hong Kong Polytechnic University, Hong Kong SAR                                                                        | 2018–2022    |
| B.Eng. in Computer Science; minor in Mathematics.                                                                          |              |

## Selected Publications

- mmHRI: Towards Privacy-Preserving Human-Robot Interaction with Millimeter-Wave Radar**  
Junqiao Fan, Yuxuan Hu, Bofan Lyu, Yanshuo Lu, Pengfei Liu, Jiarui Zhang, Fangqiang Ding, Lihua Xie, Gen Li, Jianfei Yang  
*arXiv:2609.34220, 2026 (preprint)*. [Paper] [Website]
- GIVE: Grounding Human Gestures in Vision-Language-Action Models**  
Pengfei Liu, Gen Li, Junqiao Fan, Boyu Ma, Jindou Jia, Yang Xiao, Jianfei Yang  
*arXiv:2606.13435, 2026 (preprint)*. [Paper] [Website]
- M4Human: A Large-Scale Multimodal mmWave Radar Benchmark for Human Mesh Reconstruction**  
Junqiao Fan, Yunjiao Zhou, Yizhuo Yang, Xinyuan Cui, Jiarui Zhang, Lihua Xie, Jianfei Yang, Chris Xiaoxuan Lu, Fangqiang Ding  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026*. [Paper] [Website]
- mmPred: Radar-based Human Motion Prediction in the Dark**  
Junqiao Fan, Haocong Rao, Jiarui Zhang, Jianfei Yang, Lihua Xie  
*AAAI Conference on Artificial Intelligence, 40(5):3777–3785, 2026*. [Paper] [Website]
- Diffusion Model Is a Good Pose Estimator from 3D RF-Vision**  
Junqiao Fan, Jianfei Yang, Yuecong Xu, Lihua Xie  
*European Conference on Computer Vision (ECCV), pp. 1–18, 2024*. [Paper] [Website]
- Generative Dataset Distillation Using Min-Max Diffusion Model**  
Junqiao Fan, Yunjiao Zhou, Min Chang Jordan Ren, Jianfei Yang  
*European Conference on Computer Vision Workshops, pp. 60–67, 2024*. [Paper]
- Federated Learning Driven Secure Internet of Medical Things**  
Junqiao Fan, Xuehe Wang, Yanxiang Guo, Xiping Hu, Bin Hu  
*IEEE Wireless Communications, 29(2):68–75, 2022*. [Paper]

## Honors & Awards

|                                                  |      |
|--------------------------------------------------|------|
| 2nd Place, Generative Track                      | 2024 |
| First Dataset Distillation Challenge, ECCV 2024. |      |

## Academic Service

**Journal Reviewer:** TMC, etc.  
**Conference Reviewer:** AAAI, ICLR, ICRA, etc.

## Skills

Python, MoCap, ROS2.